

WHAT IS ENGINEERING MANAGEMENT?

- Engineering management refers to the activity combining "technical knowledge with the ability to organize and coordinate worker power, materials, machinery, and money. When the engineer is assigned to supervise the work of even a few people, he is already engaged in the first phase of engineering management.
- His main responsibility is to lead his group to produce a certain output consistent with the required specifications. The top position an engineer manager may hope to occupy is the general managership or presidency of any firm, large or small. As he scales the management ladder, he finds that the higher he goes up, the less technical activities he performs, and the more management tasks he accepts.

MANAGEMENT DEFINED

- Since the Engineering Head is presumed to be technically competent in his specialization, one may now proceed to describe more thoroughly the remaining portion of his job, which is management. Management may be defined as the
- creative problem-solving process of planning, organizing, leading, and controlling an organization's resources to achieve its mission and objectives.

THE PROCESS OF MANAGEMENT

- Management is a process consisting of planning, organizing, directing (or leading), and controlling. Explained simply, management must seek to find out the objectives of the organization, think of ways how to achieve them, decide on the ways to be adapted and the material and resources to be used, determine the human requirements of the total job, assign specific tasks to specific persons, motivate them, and provide means to make sure that the activities are in the right decision.

HOW TO BECOME A SUCCESSFUL ENGINEERING MANAGER

- Successful engineering managers do not happen as a matter of chance, although luck is a contributory factor. The engineering manager needs to know the various factors leading to successful management.
- Kreitner indicates at least three general preconditions for achieving lasting success as a manager. They are as follows:
 - 1. ability
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- 2. motivation to manage, and
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- 3. opportunity.

ABILITY

- Managerial ability refers to the capacity of an engineer manager to achieve organizational objectives effectively and efficiently. Effectiveness, according to Higgins, refers to a description of "whether objectives are accomplished", while efficiency is a description of the relative amount of resources used in obtaining effectiveness.

MOTIVATION TO CHANGE

- Many people have the desire to work and finish specific tasks assigned by superiors, but not many are motivated to manage other people so that they may contribute to the realization of the organization's objectives. A management researcher, John B. Miner, developed a psychometric instrument to measure objectively an individual's motivation to manage. The test is anchored to the following dimensions:
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 - 1. Favorable attitude toward those in positions of authority, such as superiors.
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 - 2. Desire to engage in games or sports competition with peers.
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 - 3. Desire to engage in occupational or work-related competition with peers.
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 - 4. Desire to assert oneself and take charge.
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 - 5. Desire to exercise power and authority over others.
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 - 6. Desire to behave in a distinctive way, which includes standing out from the crowd.
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7. Sense of responsibility in carrying out the routine duties associated with managerial work. High scores in the foregoing dimensions are associated with high motivation to manage.

OPPORTUNITY

- Opportunity Successful managers become possible only if those having the ability and motivation are given the opportunity to manage. The opportunity for successful management has two requirements:
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 - 1. Obtaining a suitable managerial job, and
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 - 2. Finding a supportive climate once on the job.

SUMMARY:

- Engineers are known for their great contributions to the development of the world's civilization. There are many areas where their presence is necessary like research, design and development, testing, manufacturing, construction, sales, consulting, government, teaching, and management. Engineers may be found contributing their share in the various levels of organization.
- Engineering activities need to be managed and engineers are sometimes placed in positions where they have to learn management skills.
- Management is concerned with planning, organizing, leading, and controlling an organization's resources to achieve its mission and objectives. There are certain qualifications required of the engineering manager. One may become a successful if the preconditions of ability, motivation to manage, and opportunity to manage are met.

What are management styles?

- Management styles describe the distinct approaches that different managers take to support and guide a team at work. All managers have the same responsibilities, but management styles describe how different types of managers specifically use their managerial toolkit to motivate team members.
- There are a huge variety of management styles. Every manager naturally leans toward one management style, even if you don't realize it. Our management styles come out through our communication preferences, our personality types, and how we show up at work. The key thing is to identify which management style you naturally use. Identifying

your go-to management style can help you gain a better understanding of how you're communicating and supporting your team—even the things you don't realize you are doing.

Types of Management Styles

- There are a wide variety of management styles, with more being defined every day. Anything that defines how you can guide your team, plan work, and connect the dots can be considered a management style. Some of these styles are top-down, while others are more hands-off. The nine different management styles include:

(Diagram representation)

- **Management styles**
 - **Autocratic** -> Persuasive, Paternalistic
 - **Democratic** -> Consultative, Transformative
 - **Laissez-faire** -> Visionary, Delegative

1. Autocratic management style

- Autocratic managers centralize the decision-making process. On these teams, the manager usually makes the majority of decisions—including what the team should focus on, what short- and long-term goals they should work toward, and which tasks and projects are associated with these initiatives. A manager with an autocratic management style will then delegate work to the team member they think is best suited for the job. This is the most top-down approach to management.
- The advantages of an autocratic management style:
 - It's easier to make quick decisions since there is only one decision-maker.
- Best for: When decisions need to be made quickly and there is little room for error. In crisis situations, the autocratic style is the way to go.

2. Persuasive management style

- One type of autocratic management is persuasive management. The persuasive management style puts decision-making totally in the hands of managers but encourages managers to share the logic and rationale behind their decisions. This helps team members feel more connected to the decision-making process.
- The advantages of a persuasive management style:

- It can help educate junior-level team members about what goes into managerial decisions.
- It builds trust between the team and the manager since the team is given an explanation for every decision.
- Best for: Managers that are leading a more inexperienced team. Under this dynamic, the manager should be the sole decision-maker based on their experience but should share explanations in order to help their team grow.

3. Paternalistic management style

- Another type of the autocratic style, paternalistic management is a form of unilateral decision-making where the team's best interests are always kept in mind. While communication is still one-sided, team members are heard and decisions are made with their wants in mind. However, there is no collaboration between the manager and the team about the decision itself.
- In this management style, managers try to create a feeling of trust within their team. By doing so, their direct reports are confident their feedback is taken to heart whenever the manager makes a decision.
- The advantages of the paternalistic style:
 - Managers are truly committed to the well-being of their team. Team member well-being is at the heart of every decision.
 - Unlike other autocratic styles, team members are actively encouraged to express their desires to their manager, even if they don't get to collaborate on the decision-making process itself.
- Best for: Organizations with strong company culture. If leadership is able to build strong bonds within the organization, team members can trust that the right decisions are being made.

4. Democratic management style

- Democratic, or participative, managers collaborate with their team members to distribute the decision-making process. They invest in their team's well-being and career development. This management style encourages creativity and employee engagement. Though they are still the person making the final decision, managers who use a democratic style encourage contributions from all team members.

- Before setting quarterly priorities, a democratic manager will make sure every team member understands the company goals and how their team's initiatives contribute to the company's yearly objectives. This type of manager will then encourage team members to suggest their own goals. The advantages of a democratic management style:
 - Team members have the context they need to succeed.
 - Teams with democratic managers often come up with more creative ideas because everyone is brainstorming together.
- Best for: Teams that are constantly looking to change and improve their processes. If you manage this type of team, it's best that everyone stays on the same page when it comes to decision-making.

5. Consultative management style

- A subset of the democratic management style, consultative managers are consistently looking for feedback from their team and trying to improve in their position. Differing from the authoritative style, consultative managers rarely make decisions alone but instead encourage collaboration from their team.
- The advantages of the consultative management style:
 - It breeds healthy communications among teams.
 - It encourages upward feedback.
- Best for: Managers who want to take on more of a mentor role. Consultative managers strive to educate and influence their team through the decision-making process.

6. Transformational management style

- Another type of democratic management is the transformational management style. Used predominantly among creative teams, transformational managers strive to create a culture that encourages adaptability, innovation, and problem-solving.
- Transformational managers are invested in the professional growth of their team and look less at short-term projects and more at long-term development. Oftentimes, this means setting lofty and challenging goals for team members.
- The advantages of a transformational management style:
 - This style works very well for motivated team members looking to grow in their careers and roles.

- It helps breathe life into more mundane projects and helps team members see the bigger picture of what they're working toward.
- Best for: Fast-moving, intrinsically-motivated teams that are always adapting to new circumstances. By integrating adaptability into their management style, transformative managers prepare their team for whatever may come their way.

7. Laissez-faire management style

- Laissez-faire managers are the most hands-off managers. Laissez-faire literally translates to "let it go" in French, and this management style leans into that relaxed, hands-off approach. Laissez-faire managers prefer to give team members nearly full autonomy, and they typically only meet with team members or host team meetings if the team requests it. Otherwise, a laissez-faire manager only checks in if something went wrong.
- The advantages of a laissez-faire management style:
 - Team members are encouraged to be creative and come up with their own ideas.
 - Self-motivated employees benefit from increased autonomy.
- Best for: Self-motivated creative teams that thrive when building their own ideas. This model works best for teams that work individually and don't require much supervision.

8. Visionary management style

- A type of laissez-faire management, the visionary management style is less concerned with the day-to-day operations and more interested in educating team members about the larger vision of the organization. Visionary managers tend to motivate and inspire rather than give instructions about day-to-day tasks.
- These managers tend to be more outgoing and charismatic. This style works well for teams that are good at managing themselves and don't require much individual attention. For less experienced team members, this may not be the most effective style of management.
- The advantages of visionary management style:
 - When done well, this style can spur creativity and out-of-the-box thinking.
 - The visionary style can encourage your team to work harder because they're truly passionate about their work.
- Best for: Innovative teams looking to disrupt an industry. In this case, a visionary leader can be exactly what a team needs to continue thinking outside the box.

9. Delegative management style

- Another type of laissez-faire management, delegative management refers to a system where managers and team members only interact when managers are assigning them tasks. From there, the manager will come in at the end of the project, review the work, and the cycle will restart.
- In this style, team members generally have the freedom to do tasks in their own way. Like teams with visionary managers, members tend to have a strong knowledge of the work and how it needs to be done.
- The advantages of the delegative style:
 - It allows team members to maximize creativity in their assignments. With little to no supervision, team members can think outside the box when completing tasks.
 - There's no risk of micromanaging in this style. As a result, team members feel autonomous in their work.
- Best for: Employees with more expertise than their manager. In cases like this, the manager should facilitate rather than support and give feedback.

MANAGEMENT QUALITIES AND RESPONSIBILITIES

- No matter what style you choose to use, it's important to understand what it means to be a manager, and what qualities of a leader you should bring to the table in order to support your team. These include:
- Connecting daily tasks to company objectives. One of the most important things you can do for your team is to make sure they have the context they need to succeed. Only 26% of knowledge workers have a very clear understanding of how their individual work relates to company goals. Without that context, team members don't have a good sense of what they should be prioritizing, and what their highest impact work is.
- Assigning and delegating work. Part of being a good manager is delegating work to the right team member. Good managers don't think, "I can do that," but instead approach a task or project with the mentality, "I know the best person for the job." As a manager, it's your responsibility to ensure that team members have enough work that engages and challenges them but that they aren't feeling overworked or burnt out.
- Providing feedback and professional development. Good managers are forward-thinkers who want to help their team members improve and get to the next level—whatever that next level looks like for them. Part of this is providing clear, constructive feedback to help your team members turn their weaknesses into strengths. Additionally, professional

development and coaching sessions can help team members identify where they want to go in their careers—and begin moving in that direction.

- Encouraging team building and collaboration. The best teams are those that work together—and the best managers are those that help their teams do just that. Good managers make time for team building opportunities. They emphasize the importance of team collaboration and open communication, so that team members feel comfortable bringing their full selves to work.

MANAGEMENT STYLES VS. LEADERSHIP STYLES

- "Management is operational; it's about setting priorities, evaluating priorities, hiring and firing decisions, compensation decisions, things like that. A leader is more of a coach, or even a spiritual guide. She is responsible for maintaining energy, keeping everyone on the team inspired and helping them grow, and for ensuring everyone is aligned in the same direction. A leader must be a point of strength and stability across changes." — Dustin Moskovitz, co-founder and CEO, Asana
- Management styles are how managers support and guide a team at work. Managers are focused on execution, organization, and planning—so management styles are primarily about how to provide clear direction on projects and support your employees at the team level.
- Leadership styles are how leaders motivate and inspire a team or company. Leaders tend to be bigger-picture thinkers—so leadership skills primarily focus on motivating team members, leading company culture, and creative problem-solving. You can use leadership styles in management, but you need to make sure your main focus is on the specific aspects of management.
- There's a lot that goes into good management, and using the right management style is just the beginning. To continue improving your management style, actively source feedback from your team so you can support them as you grow.
- Part of this growth also means communicating openly and honestly—even when sharing constructive feedback. By proactively encouraging open and honest communication, you can ensure your team is best equipped and supported at work.

MANAGERIAL ROLES

- Managerial roles are behaviors adopted to perform various management functions, like leading and planning, organizing, strategizing, and solving problems. Within an organization, managers of different levels have different responsibilities that may overlap.

- Managerial roles are a way of classifying the various functions of management. The roles were developed by a Canadian academic and author, Professor Henry Mintzberg, and published in 1989 in his book, "Mintzberg on Management: Inside Our Strange World of Organizations."
- Mintzberg's management roles help define the skills required by managers at different levels of the organizational structure.

COMMON MANAGERIAL ROLES AND RESPONSIBILITIES

- **Interpersonal roles**
 - In interpersonal roles, the manager is responsible for providing ideas and information. They interact with other people to make sure things get done in alignment with organizational goals. This includes communication both within the organization and externally—for example, with clients and other stakeholders.
 - Interpersonal managerial roles: Figurehead, Leader, and Liaison
- **Informational roles**
 - In this category, the manager is responsible for generating or collecting data, then processing and sharing information with others. Data can be collected from inside or outside the business and it's the manager's job to analyze it and ensure it's delivered to those who need it.
 - Informational managerial roles: Monitor, Disseminator, Spokesperson
- **Decisional roles**
 - In these managerial roles, the manager uses the knowledge gathered, together with decision-making and communication, to move the organization towards its goals.
 - Decisional managerial roles: Entrepreneur, Disturbance-Handler, Resource Allocator, and Negotiator

MANAGERIAL ROLES ACCORDING TO HENRY MINTZBERG

- **1. Figurehead**
 - The figurehead role is associated with upper management and is usually invested with a fair amount of authority and power, often in tandem with legal responsibilities.
 - At this level, managers represent and promote the company externally at client meetings, legal proceedings, or social events. They have an internal role to play too,

serving as a role model and inspiring team members to keep moving towards business goals.

- **2. Leader**

- In a leadership role, managers oversee the performance of an organization, department, or team and ensure team dynamics remain positive. They set goals—like a sales or project goal—and communicate those to all team members.
- Leaders delegate tasks, check in on progress, and provide feedback, support, and motivation wherever it's needed. The leader role may also include some human resources functions, like hiring, training, and evaluating staff.

- **3. Liaison**

- The liaison role is responsible for developing and maintaining internal and external relationships. They serve as a bridge between team members in different teams or at different levels of the chain of command, communicating and connecting people to get work done.
- Liaisons may also coordinate with external stakeholders, such as meeting with clients to understand their needs or networking with partners and suppliers to further the company's goals.

- **4. Monitor**

- In a monitor role, managers actively search for information to assess the business's effectiveness and identify problems and opportunities. For example, they may gather client feedback and discover a recurring problem that requires corrective action—or get inspiration for new products.
- Monitors also keep track of industry trends to make sure the business remains competitive and stays in alignment with regulatory changes.

- **5. Disseminator**

- The function of this managerial role is to collect strategic and technical information from different sources and convey it to the appropriate people. The information can include issues raised during one-on-one or team meetings or ideas for new products or services.
- Information could also be proposals to optimize processes by implementing new operations management software. Initiatives may need approval by upper management or dissemination to employees. This can be done verbally or in writing, and either directly with the relevant person or up or down through the chain of command.

- **6. Spokesperson**

- In the spokesperson role, a manager is the face of their team or their entire organization. For example, they may present the team's results and business goals for the next year to the upper management team.
- A spokesperson might also share the company's results at a shareholder's meeting or speak at a conference to boost the company's reputation with current or potential clients and investors.

- **7. Resource Allocator**

- In the resource allocator role, the manager decides how to best allocate organizational resources to make the most of available time and budget. Examples of resources include staff members, facilities, materials, and equipment.
- The allocation may require scheduling resources according to project workload, dividing funding between departments based on budgets, or assigning equipment as it's needed.

- **10. Negotiator** (*Note: The slide skips to 10 here*)

- The negotiator role involves negotiating on behalf of a department or the entire business to achieve the best possible outcome for all concerned. For example, the manager may need to negotiate with an employee over their salary, with management over the department's budget, or with another department to gain access to team members with specific skills. Or they may negotiate with clients or vendors around deliverables, due dates, and costs.

(Diagram representation)

- **The Managerial Roles**

- **Feedback**
- **Interpersonal** (Figurehead, Leader, Liaison) -> Provide Information
- **Informational** (Monitor, Disseminator, Spokesperson) -> Process Information
- **Decisional** (Entrepreneur, Disturbance Handler, Resource Allocator, Negotiator) -> Use Information

Management Skills

- A list of attributes or abilities that an executive should possess in order to fulfill specific tasks in an organization.

What are Management Skills?

- Management skills can be defined as certain attributes or abilities that an executive should possess in order to fulfill specific tasks in an organization. They include the capacity to perform executive duties in an organization while avoiding crisis situations and promptly solving problems when they occur.
- Management skills can be developed through learning and practical experience as a manager. The skills help the manager to relate with their fellow co-workers and know how to deal well with their subordinates, which allows for the easy flow of activities in the organization
- Good management skills are vital for any organization to succeed and achieve its goals and objectives. A manager who fosters good management skills is able to propel the company's mission and vision or business goals forward with fewer hurdles and objections from internal and external sources.
- Management and leadership skills are often used interchangeably as they both involve planning, decision-making, problem-solving, communication, delegation, and time management. Good managers are almost always good leaders as well.
- In addition to leading, a critical role of a manager is to also ensure that all parts of the organization are functioning cohesively. Without such integration, several issues can arise and failure is bound to happen. Management skills are crucial for various positions and at different levels of a company, from top leadership to intermediate supervisors to first-level managers.

Types of Management Skills

- According to American social and organizational psychologist Robert Katz, the three basic types of management skills include:
- **1. Technical Skills**
- Technical skills involve skills that give the managers the ability and the knowledge to use a variety of techniques to achieve their objectives. These skills not only involve operating machines and software, production tools, and pieces of equipment but also the skills needed to boost sales, design different types of products and services, and market the services and the products.
- **2. Conceptual Skills**
- These involve the skills managers present in terms of the knowledge and ability for abstract thinking and formulating ideas. The manager is able to see an entire concept,

analyze and diagnose a problem, and find creative solutions. This helps the manager to effectively predict hurdles their department or the business as a whole may face.

- **3. Human or Interpersonal Skills**

- The human or the interpersonal skills are the skills that present the managers' ability to interact, work or relate effectively with people. These skills enable the managers to make use of human potential in the company and motivate the employees for better results.

(Diagram representation)

- **MANAGEMENT SKILLS**

- Technical Skills
- Interpersonal Skills
- Conceptual Skills *(Arranged in a cyclical workflow)*

Examples of Management Skills

- There is a wide range of skills that management should possess to run an organization effectively and efficiently. The following are six essential management skills that any manager ought to possess for them to perform their duties:
- **1. Planning**
- Planning is a vital aspect within an organization. It refers to one's ability to organize activities in line with set guidelines while still remaining within the limits of the available resources such as time, money, and labor. It is also the process of formulating a set of actions or one or more strategies to pursue and achieve certain goals or objectives with the available resources.
- The planning process includes identifying and setting achievable goals, developing necessary strategies, and outlining the tasks and schedules on how to achieve the set goals. Without a good plan, little can be achieved.

THREE STAGES OF PROJECT MANAGEMENT

- The first two stages project initiation and project planning. At this point, you know the final three phases of project management (execution, monitoring and project close) and your project management event is complete.
- The first phase in any project management process is Project Initiation. The second phase is Project Planning. Together the first two phases represent the seven "Ps" of

planning: Proper Prior Planning Prevents a Pretty Poor Program. But you are not preparing planning for planning's sake, you need the deliverables, which are in the project execution phase, followed by Project Monitoring & Control, and Project Review & Close.

- **2. Communication**

- Possessing great communication skills is crucial for a manager. It can determine how well information is shared throughout a team, ensuring that the group acts as a unified workforce. How well a manager communicates with the rest of his/her team also determines how well outlined procedures can be followed, how well the tasks and activities can be completed, and thus, how successful an organization will be.
- Communication involves the flow of information within the organization, whether formal or informal, verbal or written, vertical or horizontal, and it facilitates the smooth functioning of the organization. Established communication channels in an organization allow the manager to collaborate with the team, prevent conflicts, and resolve issues as they arise. A manager with good communication skills can relate well with the employees and, thus, be able to achieve the company's set goals and objectives easily.

- **3. Decision-making**

- Another vital management skill is decision-making. Managers make numerous decisions, whether knowingly or not, and making decisions is a key component in a manager's success. Making proper and right decisions results in the success of the organization, while poor or bad decisions may lead to failure or poor performance.
- For the organization to run effectively and smoothly, clear and right decisions should be made. A manager must be accountable for every decision that they make and also be willing to take responsibility for the results of their decisions. A good manager needs to possess great decision-making skills, as it often dictates his/her success in achieving organizational objectives.

- **4. Delegation**

- Delegation is another key management skill. Delegation is the act of passing on work-related tasks and/or authorities to other employees or subordinates. It involves the process of allowing your tasks or those of your employees to be reassigned or reallocated to other employees depending on current workloads. A manager with good delegation skills is able to effectively and efficiently reassign tasks and give authority to the right employees. When delegation is carried out effectively, it helps facilitate efficient task completion.

- Delegation helps the manager to avoid wastage of time, optimizes productivity, and ensures responsibility and accountability on the part of employees. Every manager must have good delegation abilities to achieve optimal results and accomplish the required productivity results.
- **5. Problem-solving**
- Problem-solving is another essential skill. A good manager must have the ability to tackle and solve the frequent problems that can arise in a typical workday. Problem-solving in management involves identifying a certain problem or situation and then finding the best way to handle the problem and get the best solution. It is the ability to sort things out even when the prevailing conditions are not right. When it is clear that a manager has great problem-solving skills, it differentiates him/her from the rest of the team and gives subordinates confidence in his/her managerial skills.
- **6. Motivating**
- The ability to motivate is another important skill in an organization. Motivation helps bring forth a desired behavior or response from the employees or certain stakeholders. There are numerous motivation tactics that managers can use, and choosing the right ones can depend on characteristics such as company and team culture, team personalities, and more. There are two primary types of motivation that a manager can use. These are intrinsic and extrinsic motivation.
- Management skills are a collection of abilities that include things such as business planning, decision-making, problem-solving, communication, delegation, and time management. While different roles and organizations require the use of various skill sets, management skills help a professional stand out and excel no matter what their level. In top management, these skills are essential to run an organization well and achieve desired business objectives.

How does good management benefit a company?

- Good management helps ensure an organization's success in several ways. These include:
 - **Planning:** Good managers plan strategies to achieve company goals.
 - **Organization:** Good managers know about company resources and how to allocate them.
 - **Direction:** Good managers know how to direct and motivate employees.

- **Control:** Good managers ensure employees execute tasks and make changes when needed.

Ways to develop managerial skills

- Gaining new skills or improving the ones you have can help you be more successful in a management position. Whether you're a current manager or hoping to become one, consider these tips for developing management skills:
 - Seek advice and guidance from a mentor.
 - Read books about management theories, techniques, and styles.
 - Think about areas you need to improve, set goals, and take steps to achieve them.
 - Go back to school and get a certificate or a degree.
 - Attend industry-related training, conferences, and workshops.
 - Practice skills like active listening, delegating, and organization.
 - Join a public speaking group or take a public speaking or business writing class.
 - Look for opportunities to be a leader at work, home, class, or through volunteering or sports.

LEVELS OF MANAGEMENT

- The term Levels of Management refers to the line of division that exists between various managerial positions in an organization. As the size of the company and workforce increases, the number of levels in management increases along with it, and vice versa. The different Levels of Management can determine the chain of command within an organization, as well as the amount of authority and typically decision-making influence accrued by all managerial positions.
- Levels of Management can be generally classified into three principal categories, all of which direct managers to perform different functions.
- In this article, we will explore the specific definition of these levels, as well as the roles and responsibilities of the managers that fall into these categories.

1. Administrative, Managerial, or Top Level of Management

- This level of management consists of an organization's board of directors and the chief executive or managing director. It is the ultimate source of power and authority since it

oversees the goals, policies, and procedures of a company. Their main priority is on the strategic planning and execution of the overall business success.

- The roles and responsibilities of the top level of management can be summarized as follows:
 - Issuing necessary instructions for the preparation of department-specific budgets, schedules, procedures, etc.
 - Preparing strategic plans and policies for the organization.
 - Appointing the executives for middle-level management, i.e. departmental managers.
 - Establishing controls of all organizational departments.
 - Since it consists of the Board of Directors, the top management level is also responsible for communicating with the outside world and is held accountable towards an organization's shareholders for the performance of the enterprise.
 - Providing overall guidance, direction, and encouraging harmony and collaboration.

2. Executive or Middle Level of Management

- The branch and departmental managers form this middle management level. These people are directly accountable to top management for the functioning of their respective departments, devoting more time to organizational and directional functions. For smaller organizations, there is often only one layer of middle management, but larger enterprises can see senior and junior levels within this middle section.
- The roles and responsibilities of the middle level of management can be summarized as follows:
 - Executing the plans of the organization in accordance with the policies and directives laid out by the top management level.
 - Forming plans for the sub-units of the organization that they supervise.
 - Participating in the hiring and training processes of lower-level management.
 - Interpreting and explaining the policies from top-level management to lower-level management.
 - Sending reports and data to top management in a timely and efficient manner.

- Evaluating the performance of junior managers.
- Inspiring lower level managers towards improving their performance.

3. Supervisory, Operative, or Lower Level of Management

- This level of management consists of supervisors, foremen, section officers, superintendents, and all other executives whose work must do largely with HR oversight and the direction of operative employees. Simply put, managers at the lower level are primarily concerned with the execution and coordination of day-to-day workflow that ensure completion of projects and that deliverables are met.
- The roles and responsibilities of the lower level of management can be summarized as follows:
 - Assigning jobs and tasks to various workers.
 - Guiding and instructing workers in day-to-day activities.
 - Maintaining good relations within lower levels of the organization.
 - Acting as mediators by communicating the problems, suggestions, and recommendatory appeals, etc. of workers to the higher level of management, and in turn elucidating higher-level goals and objectives to workers.
 - Helping to address and resolve the grievances of workers.
 - Supervising and guiding their subordinates.
 - Taking part in the hiring and training processes of their workers.
 - Arranging the necessary materials, machines, tools, and resources, etc. necessary for accomplishing organizational tasks.
 - Preparing periodical reports regarding the performance of the workers.
 - Upholding discipline, decorum, and harmony within the workplace.

Levels of Management Summary

- An organization can have many different managers, across a variety of titles, authority levels, and levels of the management hierarchy that we illustrated above. In order to properly assign roles and responsibilities to all managerial positions, it is important to recognize the key differences between low-level, middle-level, and top-level management.
- The key takeaways from this distinction are as follows:

- Top-level managers are responsible for controlling and overseeing the entire organization.
- Middle-level managers are responsible for executing organizational plans which comply with the company's policies. They act as an intermediary between top-level and low-level management.
- Low-level managers focus on the execution of tasks and deliverables, serving as role models for the employees they supervise.
- All businesses are comprised of a vast array of different managerial tasks. When these are coordinated properly, and there is a strong hierarchical manager system in place, an organization can be extremely efficient in creating value through the production of their products, services and overall workflow.

Engineering Manager Skills And Characteristics

- Examples of skills that engineering managers should have include high-level technical knowledge (from both a degree and experience), project management skills (like organization), problem-solving, emotional intelligence, and communication. All skills that managers should have fall into three main categories: human, technical, and hybrid.
- These categories encompass an assortment of skill sets and characteristics, such as:
 - **Human skills:** Leadership, accountability, innovation, emotional intelligence, and motivation
 - **Hybrid skills:** Communication, problem-solving, and project management
 - **Technical skills:** Operations, research, troubleshooting, and integration

HUMAN SKILLS

- Human skills are arguably the most meaningful engineering manager competencies, and they relate to how you interact with your team and leverage your position. Although your team is pursuing technical objectives, you cannot be an effective team member or leader if you aren't able to connect with your team and maintain positive relationships.
- **Accountability:** Holding yourself accountable for your actions and their consequences – whether intended or not – is the mark of a great leader. Accepting responsibility for situations you were part of will help you grow from mistakes and become part of the solution, and it will earn you the respect of your peers.
- **Emotional intelligence:** Emotional intelligence refers to being able to identify, respond to and manage your emotions as well as others' emotions. It allows you to interact with

others in a positive, calm manner that's mindful of their needs, helping you to de-escalate conflicts, identify when peers need support, and generate positive camaraderie amongst the team.

- **Innovation:** Developing creative solutions or approaches to project problems, technical processes, or team conflicts will help you and your team succeed. Being innovative when faced with a challenge is a strength that can help to avoid delays, encourage others, and find unique solutions for the situation at hand.
- **Leadership:** Being able to step up and take the lead when it's needed can prevent mistakes, accidents, and project delays. You should be able to guide others and offer support or advice when you know your experience could be useful. The code of ethics for engineers states that they are expected to exhibit the highest standards of honesty and integrity, since engineering has a vital impact on quality of life.
- **Motivation/influence:** The ability to motivate and influence others will help you improve morale, explain your decisions, and get everyone on the same page when it comes to project processes or goals. It can create a better work environment with few project delays and a stronger team mindset.

TECHNICAL SKILLS

- Technical skills are directly related to the job and project-oriented tasks. They're essential for performing tasks efficiently and strategically, minimizing costs and delays while optimizing productivity.
- **Integration:** Systems integration – designing, testing, implementing, and integrating a new system and its components – relies on engineering expertise and current knowledge of codes and processes. It's an overarching skill that's integral to most engineering organizations, and having a thorough understanding of it will aid in tasks like troubleshooting and process development.
- **Operations:** You need to have the technical skills required to maintain operations, including staying up to date on programming languages, algorithms, frameworks, cloud services, tools, and methodologies. As a team member, you'll use these technical skills every day, and as a leader, you'll need a firm understanding of them to step in where needed and inform project decisions.
- **Troubleshooting:** When something malfunctions – be it an application or machinery – you should be able to step in and troubleshoot, leveraging your experience and skills to identify and solve the problem. While some issues may be outside of your scope, many

issues can be resolved through troubleshooting, minimizing downtime and making it a necessary skill to have.

HYBRID SKILLS

- Hybrid skills blend technical knowledge with human knowledge and soft skills. They involve the management of both the project and the team to maintain successful and optimized operations and working conditions.

RESPONSIBILITY VS AUTHORITY VS ACCOUNTABILITY

- Responsibility, authority and accountability go with your job as a supervisor.
- Responsibility includes the duties and tasks which you carry because of your position or function.
- Authority is the power and rights you are given, so you can effectively discharge your responsibility.
- **ACCOUNTABILITY** - refers to your liability or answerability as supervisor for the way in which your organizational obligations and functions are discharged either by yourself or those whom you have to delegate it.
- Thus, if anything goes wrong in your area of responsibility, you should not attempt to assign blame to anyone.
- This is the concept of command responsibility.